

Team 7 - E-Waste Battery Extraction

CV-23: Downstream Detection Consumer Integration

Full Technical Implementation Report

ROS 2 Jazzy, Gazebo, MoveIt, Kinova and cross-component perception integration

Ubuntu 24.04 on WSL2

Senior Capstone - Sprint 2 Technical Contribution

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1 Task Overview

CV-23 required Team 7 to create or connect a downstream ROS 2 consumer that receives the agreed CV detection output and demonstrates that its centroid or world-coordinate information can be used correctly by planning and robotics components.

The final contribution goes beyond printing a detection message. It implements and validates the following chain:

`/phone_detections` → CV-23 consumer → `/cv_target_pose` → MoveIt IK → Kinova pick-place
→ Gazebo state verification

The work was carried out in two practical phases. Part A established the downstream ROS 2 consumer and planning-friendly pose interface. Part B connected that interface to the existing robot script, added safety checks, performed dry-run IK validation, and executed a controlled Gazebo pick-place sequence.

2 System and Integration Context

The project was run in Ubuntu 24.04 under WSL2 with ROS 2 Jazzy, Gazebo and MoveIt. The main project path was:

```
/mnt/e/team7_project_bundle
```

CV-23 depended on earlier perception work rather than replacing it. The detector already published a frame-level custom message on `/phone_detections`, while the localisation integration populated explicit world-coordinate fields. The downstream task was therefore able to consume a stable interface instead of reopening camera geometry inside the robot layer.

A five-terminal convention was used during the main integration test:

Terminal	Process	Purpose
1	Gazebo simulation	world, camera and robot
2	localisation TF	camera-to-world transform
3	ONNX detector	detection and localisation output
4	CV-23 consumer	detection-to-pose adapter
5	checks / robot script	topic inspection, IK and execution

3 Input Detection Contract

The installed `PhoneDetection` message was inspected directly before implementation. It contained the expected class, confidence, pixel geometry and projected world position:

```
string class_id
float64 score
int32 bbox_x1, bbox_y1, bbox_x2, bbox_y2
float64 mar_center_x, mar_center_y
float64 world_x, world_y, world_z
bool world_valid
```

```

ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 interface show team7_msgs/msg/PhoneDetection
std_msgs/Header header
  builtin_interfaces/Time stamp
    int32 sec
    uint32 nanosec
  string frame_id
string class_id          # from the model's class map, e.g. "battery"
float64 score            # 0..1

# pixel-space, in the ORIGINAL /top_camera/image resolution (320x240)
int32 bbox_x1
int32 bbox_y1
int32 bbox_x2
int32 bbox_y2

# oriented box from the segmentation mask (cv2.minAreaRect), pixel-space
float64 mar_center_x
float64 mar_center_y
float64 mar_width
float64 mar_height
float64 angle_deg

# projected world-frame position (ground-plane assumption, see geometry.py)
float64 world_x
float64 world_y
float64 world_z
bool    world_valid      # false if camera_info not yet received, etc.

```

Figure 1: Installed CV detection message used as the downstream input contract.

The frame-level `PhoneDetectionArray` groups all detections from one processed image. This allows a consumer to select among candidates while retaining a clear empty-array representation when nothing is detected.

4 Part A - Downstream Consumer Implementation

4.1 Node Responsibilities

A new node `detection_consumer_node.py` was added to the perception package. Its responsibility is intentionally narrow:

1. receive `/phone_detections`;
2. keep detections matching the configured class;
3. enforce a configurable minimum confidence;
4. require `world_valid=true`;
5. select the highest-confidence valid candidate;
6. publish that position on `/cv_target_pose` as `PoseStamped`.

The default parameters were:

```

input_topic   = /phone_detections
output_topic  = /cv_target_pose
target_class  = battery
min_score     = 0.40

```

The output carries the source timestamp, explicitly sets `frame_id='world'`, copies the world X/Y/Z position, and uses identity orientation as a placeholder because the current CV message does not estimate a robot grasp orientation.

4.2 Packaging

The node was registered as a normal ROS 2 console script and `geometry_msgs` was added as a package dependency. The file passed Python syntax checking and the rebuilt package exposed both

executables:

```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 pkg executables team7_perception
team7_perception detection_consumer
team7_perception onnx_phone_detector
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$
```

Figure 2: ROS 2 sees the original ONNX detector and the new CV-23 detection consumer.

5 Part A - Live Interface Validation

The consumer was run alongside Gazebo, localisation TF and the ONNX detector. It received a live battery-class message with score approximately 0.907 and published the corresponding world pose.

```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 run team7_perception detection_consumer
[INFO] [1789887206.827973496]: [detection_consumer]: CV-23 consumer started: /phone_detections -> /cv_target_pose
[INFO] [1789887206.868668186]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887206.924776751]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887207.383387131]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887208.512921607]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887209.545409519]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887210.487821331]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887211.349084230]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887212.436982126]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887213.263720506]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887214.270974610]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887215.364510517]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887216.398756221]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887217.699286615]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887218.814963995]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887220.126134983]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887221.453642730]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887222.628290369]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887223.720212240]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887224.886004357]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887226.181676544]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887227.435449445]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887228.394136841]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887229.395160134]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887230.578781471]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887231.5908531398]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887232.609649165]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887233.697935951]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887234.576495650]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887235.908544831]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887236.808750213]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
```

Figure 3: Live consumer callback accepting the upstream battery detection.

The relevant source values were:

Quantity	Value
MAR centre	(1078.5, 464.5) px
World X	0.015797 m
World Y	0.153102 m
World Z	0.770000 m
World valid	true

The downstream PoseStamped contained the same position and declared the world frame:

```

ad@DESKTOP-OT45K60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 topic echo /cv_target_pose --once
header:
  stamp:
    sec: 352
    nanosec: 0
  frame_id: world
pose:
  position:
    x: 0.01579691140254362
    y: 0.15310204041751047
    z: 0.77
  orientation:
    x: 0.0
    y: 0.0
    z: 0.0
    w: 1.0
---
```

Figure 4: Live /cv_target_pose produced from the accepted CV detection.

This completes the perception-to-planning interface portion of CV-23.

6 Part A - Invalid Input Behaviour

A high-confidence battery message with `world_valid=false` was injected on an isolated test topic. The consumer correctly logged `No valid downstream target in this frame.` and produced no accepted target.

```

ad@DESKTOP-OT45K60:~$ ros2 run team7_perception detection_consumer --ros-args \
-p input_topic:=/cv23_test_detections \
-p output_topic:=/cv23_test_target
[INFO] [1789811670.937749451] [detection_consumer]: CV-23 consumer started: /cv23_test_detections -> /cv23_test_target
[INFO] [1789811670.999549986] [detection_consumer]: Target class="battery", min_score=0.40
[INFO] [1789811698.002776368] [detection_consumer]: No valid downstream target in this frame.
```

Figure 5: Consumer-level negative test for invalid world localisation.

This behaviour prevents downstream nodes from interpreting placeholder numeric coordinates as a usable robot target.

7 Part B - Existing Planning and Robot Interface

The supplied simulation already contained a complete pick-place script. Its intended perception hook was explicit: the battery XY position was hard-coded as `(-0.15, 0.62)` and later assigned to `bx`, `by` before building the pick-place plan.

The script already used:

- `geometry_msgs/PoseStamped`;
- `MoveIt /compute_ik` and `/compute_fk`;
- `/joint_trajectory_controller/joint_trajectory`;
- a downward-facing suction tool orientation;
- Gazebo model-state queries for final validation.

CV-23 therefore changed the source of the target rather than rewriting the robot controller.

8 Part B - CV-Aware Pick-Place Consumer

The original script was preserved and two copies were created:

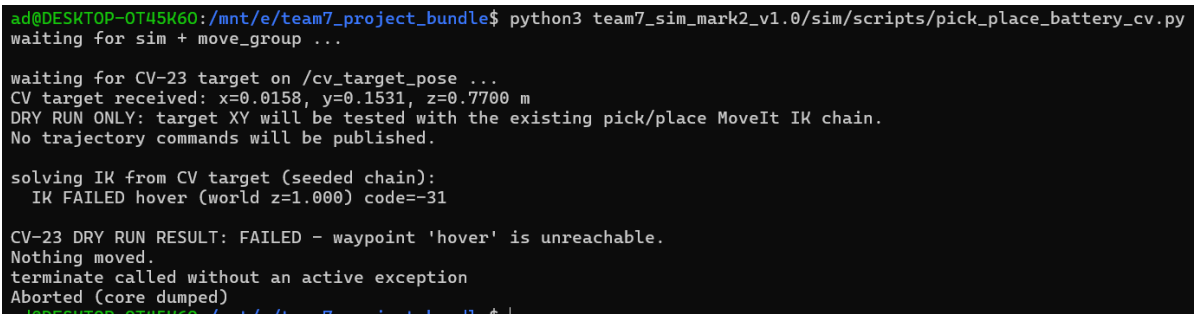
```
pick_place_battery_cv.py
pick_place_battery_cv_execute.py
```

Each copy subscribes to `/cv_target_pose`, waits for one world-frame target, and uses its X/Y values when constructing the hover, contact and lift waypoints. The drop-box waypoints remain fixed scene targets.

The dry-run version contains no trajectory or suction execution after planning. This allowed the real CV pose to be tested through the same MoveIt service chain with zero arm movement.

9 Live CV Pose Through MoveIt

The real CV pose from the live detector was approximately (0.0158, 0.1531, 0.7700). MoveIt rejected the first hover waypoint with error code -31 and the script aborted before any movement.



```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$ python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=0.0158, y=0.1531, z=0.7700 m
DRY RUN ONLY: target XY will be tested with the existing pick/place MoveIt IK chain.
No trajectory commands will be published.

solving IK from CV target (seeded chain):
  IK FAILED hover (world z=1.000) code=-31

CV-23 DRY RUN RESULT: FAILED - waypoint 'hover' is unreachable.
Nothing moved.
terminate called without an active exception
Aborted (core dumped)
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$
```

Figure 6: Live detector target reaches the downstream MoveIt consumer but fails the first IK waypoint.

This failure was consistent with the known upstream detector issue: the reported battery bounding box covered a very large portion of the image and its centroid did not correspond to the actual `battery_1` model. CV-23 therefore treated detector quality and downstream integration as separate concerns.

10 Controlled Downstream IK Validation

A known-good target at (-0.15, 0.62, 0.80) was then published through the same ROS interface. The dry-run solved all six waypoints successfully.

```

ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$ python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=-0.1500, y=0.6200, z=0.8000 m
DRY RUN ONLY: target XY will be tested with the existing pick/place MoveIt IK chain.
No trajectory commands will be published.

solving IK from CV target (seeded chain):
hover      target=(-0.1500, 0.6200, 1.000) jump=2.82 rad
contact    target=(-0.1500, 0.6200, 0.855) jump=0.27 rad
lift       target=(-0.1500, 0.6200, 1.070) jump=0.43 rad
over-box   target=(0.6200, 0.1500, 0.700) jump=1.66 rad
release    target=(0.6200, 0.1500, 0.550) jump=0.42 rad
retreat    target=(0.6200, 0.1500, 0.700) jump=0.42 rad

CV-23 DRY RUN RESULT: PASSED
All downstream pick/place waypoints have valid IK solutions.
The target came from /cv_target_pose.
NO ROBOT MOTION WAS EXECUTED.
terminate called without an active exception
Aborted (core dumped)

```

Figure 7: Controlled CV target passes all six MoveIt IK waypoints without robot motion.

The successful chain included hover, contact, lift, over-box, release and retreat. Each solution was seeded from the previous waypoint to avoid discontinuous solutions on the 7-DOF arm.

11 Execution Safety Design

The execution version performs two checks before the first trajectory command:

1. all six IK waypoints must be valid;
2. for the controlled integration experiment, the target must lie within 3 cm of the verified battery XY position.

This makes the permission to move stricter than simple geometric reachability. A separate preflight also confirmed that the controlled publisher was the only publisher on `/cv_target_pose` before the arm was allowed to execute.

12 Controlled End-to-End Robot Execution

The controlled target passed both the IK chain and the safety gate. The existing robot sequence was then executed through the normal joint trajectory controller and simulated suction mechanism.

```

ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$ python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv_execute.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=-0.1500, y=0.6200, z=0.8000 m
DRY RUN ONLY: target XY will be tested with the existing pick/place MoveIt IK chain.
No trajectory commands will be published.

solving IK from CV target (seeded chain):
hover      target=(-0.1500, 0.6200, 1.000) jump=2.82 rad
contact    target=(-0.1500, 0.6200, 0.855) jump=0.27 rad
lift       target=(-0.1500, 0.6200, 1.070) jump=0.43 rad
over-box   target=(0.6200, 0.1500, 0.700) jump=1.66 rad
release    target=(0.6200, 0.1500, 0.550) jump=0.42 rad
retreat    target=(0.6200, 0.1500, 0.700) jump=0.42 rad

CV-23 IK CHECK: PASSED
All downstream pick/place waypoints have valid IK solutions.

SAFETY CHECK: PASSED
Executing controlled CV-23 pick/place sequence...
battery before: [-0.149998, 0.619999, 0.8]
  hover over CV target: 2.82 rad / 5.6s sim (<= 226s wall)
    reached in 11s
  dip to contact: 0.25 rad / 1.0s sim (<= 90s wall)
    reached in 2s
  SUCTION ON - battery sticks to pad
  lift: 0.43 rad / 1.0s sim (<= 90s wall)
    reached in 3s
  traverse to box: 1.66 rad / 3.3s sim (<= 133s wall)
    reached in 9s
  lower over box: 0.42 rad / 1.0s sim (<= 90s wall)
    reached in 4s
  SUCTION OFF - battery drops
  retreat: 0.42 rad / 1.0s sim (<= 90s wall)
    reached in 2s

battery before: [-0.149998, 0.619999, 0.8]
battery after : [0.707647, 0.112445, 0.039959]
CV-23 EXECUTION RESULT: PASSED - battery is IN THE BOX

```

Figure 8: Successful controlled downstream execution from ROS target through MoveIt and the simulated Kinova arm.

The evidence screenshot still contains an inherited “DRY RUN ONLY” banner from the earlier copy of the script. This wording was cleaned up immediately afterwards; it was a message-only issue and did not reflect the execution path shown in the same terminal output, where trajectory and suction commands were actually run.

The main terminal result was:

```

CV-23 IK CHECK: PASSED
SAFETY CHECK: PASSED
...
CV-23 EXECUTION RESULT: PASSED - battery is IN THE BOX

```

The model state before execution was approximately $[-0.149998, 0.619999, 0.8]$. After release and retreat, the battery state was approximately $[0.707647, 0.112445, 0.039959]$.

13 Gazebo Ground-Truth Check

An independent Gazebo query after execution reported:

Model	X (m)	Y (m)	Z (m)
battery_1	0.707647	0.112445	0.039960
drop_box	0.620000	0.150000	0.000000

The battery’s final XY distance from the box centre is approximately 0.095 m and its Z coordinate is approximately 0.040 m, satisfying the existing box acceptance condition.

```

ad@DESKTOP-OT45K60:~$ gz model -m battery_1 -p

Requesting state for world [ewaste_room]...

Model: [28]
- Name: battery_1
- Pose [ XYZ (m) ] [ RPY (rad) ]:
  [0.707647 0.112445 0.039960]
  [1.567650 -0.828826 1.759420]
ad@DESKTOP-OT45K60:~$ gz model -m drop_box -p

Requesting state for world [ewaste_room]...

Model: [40]
- Name: drop_box
- Pose [ XYZ (m) ] [ RPY (rad) ]:
  [0.620000 0.150000 0.000000]
  [0.000000 -0.000000 0.000000]

```

Figure 9: Independent Gazebo model poses after the successful pick-place run.

14 Negative Safety Test

A deliberately shifted target $(-0.15, 0.67, 0.80)$ was used to test the execution gate. All six IK waypoints were valid, proving the target was robot-reachable. The 3 cm validation tolerance then rejected it before any `pp.go()` call.

```

ad@DESKTOP-OT45K60:/mnt/e/team7_project_bundle$ cd /mnt/e/team7_project_bundle
python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv_execute.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=-0.1500, y=0.6700, z=0.8000 m
CV-23 target will first be validated with the existing pick/place MoveIt IK chain.
Robot motion will only begin after both IK and the safety check pass.

solving IK from CV target (seeded chain):
hover      target=(-0.1500, 0.6700, 1.000) jump=2.74 rad
contact    target=(-0.1500, 0.6700, 0.855) jump=0.26 rad
lift       target=(-0.1500, 0.6700, 1.070) jump=0.45 rad
over-box   target=(0.6200, 0.1500, 0.700) jump=4.43 rad
release    target=(0.6200, 0.1500, 0.550) jump=0.42 rad
retreat    target=(0.6200, 0.1500, 0.700) jump=0.42 rad

CV-23 IK CHECK: PASSED
All downstream pick/place waypoints have valid IK solutions.
SAFETY ABORT: received target is not within 3 cm of the controlled validation position.
Received (-0.1500, 0.6700); expected approximately (-0.1500, 0.6200).
Nothing moved.
terminate called without an active exception
Aborted (core dumped)

```

Figure 10: Reachable-but-unapproved target passes IK but is rejected by the CV-23 safety gate.

The distinction is important:

IK success alone does not authorise motion.

15 Debugging and Engineering Corrections

15.1 Missing localisation launch file in the E-drive copy

The E-drive workspace initially lacked `localisation_tf.launch.py`. The tested file was recovered from the completed WSL-home CV-17 workspace, copied into the active package, and added to `setup.py` so the launch file was installed correctly.

15.2 Robot-script shutdown abort

The first dry-run and negative-test versions exited after the expected result but produced a C++ termination/core-dump message. The cause was the Python process finishing while the background ROS spin thread was still alive. A cleanup function now calls `rclpy.shutdown()`, joins the thread and destroys the node.

```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle$ cd /mnt/e/team7_project_bundle

python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv_execute.py
waiting for sim + move_group ...

waiting for CV-23 target on /cv_target_pose ...
CV target received: x=-0.1500, y=0.6700, z=0.8000 m
CV-23 target will first be validated with the existing pick/place MoveIt IK chain.
Robot motion will only begin after both IK and the safety check pass.

solving IK from CV target (seeded chain):
  hover    target=(-0.1500, 0.6700, 1.000) jump=2.74 rad
  contact  target=(-0.1500, 0.6700, 0.855) jump=0.26 rad
  lift     target=(-0.1500, 0.6700, 1.070) jump=0.45 rad
  over-box target=(0.6200, 0.1500, 0.700) jump=2.90 rad
  release  target=(0.6200, 0.1500, 0.550) jump=0.42 rad
  retreat  target=(0.6200, 0.1500, 0.700) jump=0.42 rad

CV-23 IK CHECK: PASSED
All downstream pick/place waypoints have valid IK solutions.
SAFETY ABORT: received target is not within 3 cm of the controlled validation position.
Received (-0.1500, 0.6700); expected approximately (-0.1500, 0.6200).
Nothing moved.
```

Figure 11: Safety-abort path after the ROS thread cleanup fix. The program returns without the previous core dump.

16 Final Artefact Bundle

A dedicated `cv23_final/` directory was created to preserve the key implementation and interface artefacts:

```
PhoneDetection_interface.txt
PhoneDetectionArray_interface.txt
detection_consumer_node.py
pick_place_battery_cv.py
pick_place_battery_cv_execute.py
```

```

ad@DESKTOP-OT45K60:/mnt/e/team7_project_bundle$ ls -lh cv23_final
total 41K
-rwxrwxrwx 1 ad ad 535 Sep 19 19:42 PhoneDetectionArray_interface.txt
-rwxrwxrwx 1 ad ad 701 Sep 19 19:42 PhoneDetection_interface.txt
-rwxrwxrwx 1 ad ad 3.3K Sep 19 19:24 detection_consumer_node.py
-rwxrwxrwx 1 ad ad 14K Sep 19 19:24 pick_place_battery_cv.py
-rwxrwxrwx 1 ad ad 15K Sep 19 19:24 pick_place_battery_cv_execute.py
ad@DESKTOP-OT45K60:/mnt/e/team7_project_bundle$ wc -l cv23_final/*interface.txt
 26 cv23_final/PhoneDetectionArray_interface.txt
 26 cv23_final/PhoneDetection_interface.txt
 52 total

```

Figure 12: Frozen CV-23 handover bundle with the consumer, robot scripts and message-interface snapshots.

17 Interpretation and Limitations

CV-23 demonstrates that the perception output can be consumed correctly by downstream ROS 2 and robot-planning components. The live detector message was successfully transformed into a world-frame planning pose, and that pose reached the existing MoveIt consumer.

The report deliberately separates this downstream result from detector accuracy. The live YOLO battery centroid remained incorrect during the integration session, so the real live target was not used to command the arm. Instead, the robot execution used a controlled known-good target through the same `/cv_target_pose` interface. This verifies the consumer, planning and execution path without hiding the upstream detector limitation.

Other current boundaries are:

- the CV interface provides target position but not a grasp orientation;
- the controlled execution safety tolerance is an integration-test guard, not a final general-purpose workspace policy;
- battery Z/contact heights still use known scene geometry;
- a future production node should define a target-selection policy for multiple valid batteries and may add target ageing/timeouts.

18 Final Technical Contribution

CV-23 produced a complete cross-component integration contribution rather than a standalone subscriber. The final work can be summarised as follows:

A ROS 2 downstream consumer was implemented for the Team 7 detection interface. It filters battery detections by class, confidence and world-coordinate validity, publishes a standard world-frame target pose, and connects that target to the existing MoveIt/Kinova pick-place component. The integration was validated with live CV messages, controlled IK tests, invalid-input rejection, a reachable-but-unsafe target test, and a successful controlled Gazebo pick-place sequence whose final battery position was independently verified from the simulator state.

The strongest evidence is the combination of three results:

1. live `/phone_detections` -> `/cv_target_pose` transfer;
2. successful six-waypoint MoveIt consumption through the CV target interface;
3. successful controlled robot execution with post-run Gazebo ground-truth verification.

A Core Reproduction Commands

A.1 Simulation and localisation

```
bash /mnt/e/team7_project_bundle/run_team7_sim.sh

cd /mnt/e/team7_project_bundle/team7_perception_ws
source /opt/ros/jazzy/setup.bash
source install/setup.bash
ros2 launch team7_perception localisation_tf.launch.py
```

A.2 Perception and CV-23 consumer

```
bash /mnt/e/team7_project_bundle/run_team7_perception.sh
ros2 run team7_perception detection_consumer
```

A.3 Live inspection

```
ros2 topic echo /phone_detections --once
ros2 topic echo /cv_target_pose --once
```

A.4 Robot validation

```
python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv.py
python3 team7_sim_mark2_v1.0/sim/scripts/pick_place_battery_cv_execute.py
```

A.5 Gazebo result check

```
gz model -m battery_1 -p
gz model -m drop_box -p
```

B Generated CV-23 Artefacts

The final handover directory contains the exact downstream consumer, the dry-run and execution robot scripts, and snapshots of both custom detection interfaces. These files, together with the terminal evidence preserved in this report, provide a reproducible record of the CV-23 implementation and validation.